

AN DHCP

DHCP Service

Section 1 — DHCP Service Brief

1.1 What is DHCP?

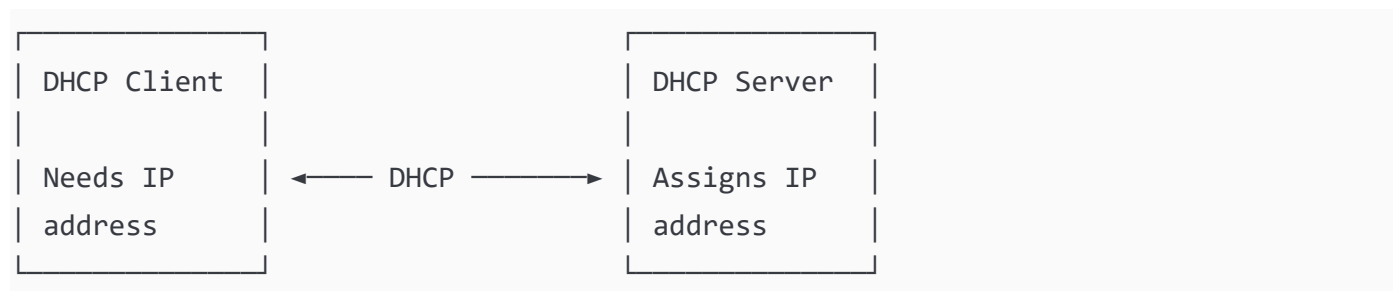
DHCP (Dynamic Host Configuration Protocol) is a network protocol used to automatically assign and manage **IP addresses and other network configuration settings** for devices on a network.

A DHCP server can provide:

- **IP Address** — Identifies the device on the network.
 - **Subnet Mask** — Defines the network and host portions of the IP address.
 - **Default Gateway** — Defines where traffic destined for other networks should be sent.
 - **DNS Server** — Provides the DNS server information used for name resolution.
 - **Lease Duration** — Defines how long the client can use the assigned IP address.
-

1.2 DHCP Client and DHCP Server

DHCP communication mainly involves two roles:



DHCP Client

The **DHCP client** is the device that requests network configuration.

Examples:

- Linux workstation
- Windows PC
- Laptop
- Virtual machine
- Printer

- Other network devices

DHCP Server

The **DHCP server** maintains the available IP address pool and provides network configuration to DHCP clients.

For our Linux lab, the DHCP server uses:

Package: `isc-dhcp-server`

1.3 DHCP Communication Ports

DHCP uses **UDP** for communication.

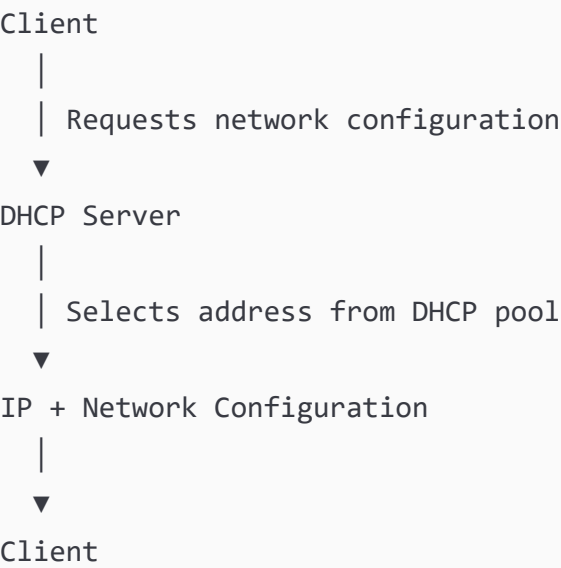
Port	Used By	Purpose
UDP 67	DHCP Server	Receives DHCP requests
UDP 68	DHCP Client	Receives DHCP responses

The initial client request uses:

Source: UDP 68
Destination: UDP 67

1.4 DHCP Configuration Flow

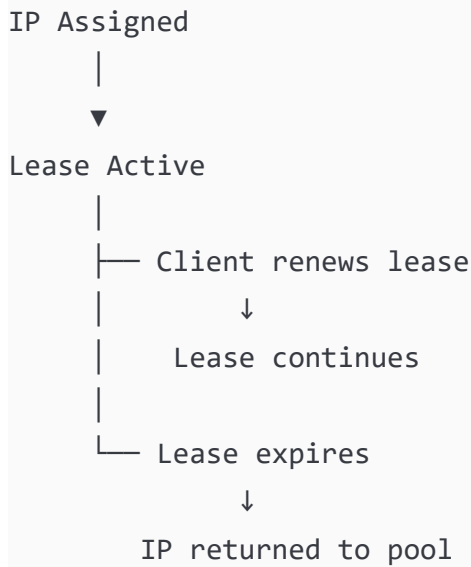
The basic operation of DHCP can be represented as:



The actual DHCP exchange is handled through the **DORA process**, which will be covered in the next section.

1.5 DHCP Lease

DHCP normally assigns an IP address for a specific **lease duration** rather than permanently assigning it.



A client can also voluntarily release its IP address when it no longer needs it.

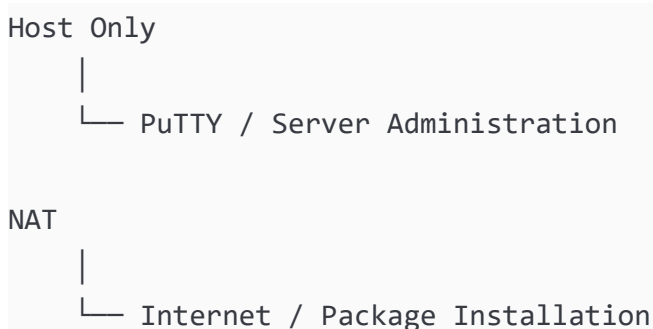
If the lease expires without renewal, the address becomes available for assignment to another client.

1.6 DHCP in Our Linux Lab

Our DHCP server will use the following network plan:

Configuration	Value
Network	192.168.100.0/24
DHCP Server IP	192.168.100.10
DHCP Range	192.168.100.101 - 192.168.100.150
DHCP Interface	enp0s9
DHCP Package	isc-dhcp-server

Network arrangement:



Internal Network



Section 2 — DHCP Architecture & DORA Process

2.1 DHCP Communication Architecture

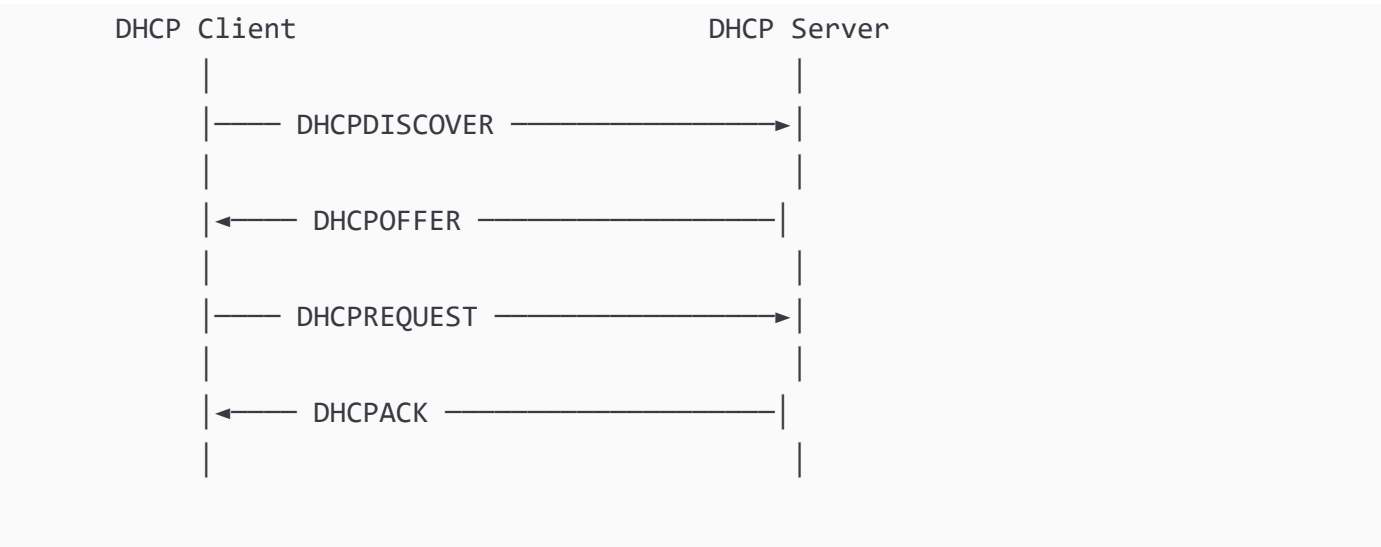


When a client joins the network without a valid IP address, it starts the DHCP process by broadcasting a **DHCPDISCOVER** message.

The server receives the request, selects an available address from its configured pool, and communicates the proposed configuration back to the client.

2.2 DORA Process

The normal DHCP address-assignment process consists of four steps:



▼
IP Assigned

DORA stands for:

D → Discover

O → Offer

R → Request

A → Acknowledgment

2.3 DHCPDISCOVER — Discover

When the DHCP client connects to the network and does not have a preconfigured IP address, it sends a **DHCPDISCOVER** broadcast to find available DHCP servers.

The initial request uses:

Source Port → UDP 68

Destination Port → UDP 67

The client does not yet know the DHCP server's IP address, so the message is broadcast across the local network.

Client

|
| DHCPDISCOVER
| Broadcast
| UDP 68 → 67



Network



DHCP Server(s)

If multiple DHCP servers are available, more than one server can receive the discovery and respond with an offer.

2.4 DHCPOFFER — Offer

A DHCP server that receives the **DHCPDISCOVER** responds with a **DHCPOFFER**.

The offer can contain:

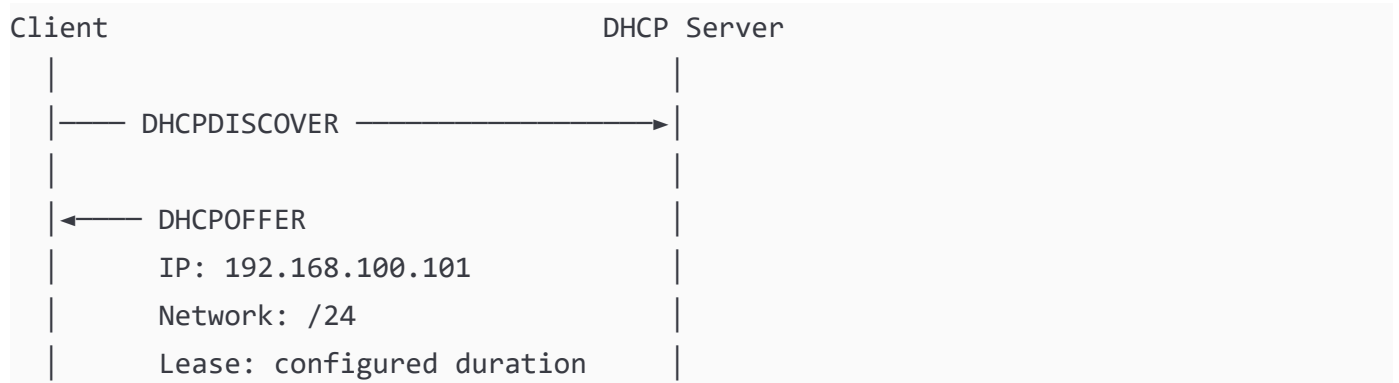
- IP address
- Subnet mask

- Lease duration
- Default gateway
- DNS server
- Other DHCP configuration parameters

For our lab, the server has the following available range:

192.168.100.101 – 192.168.100.150

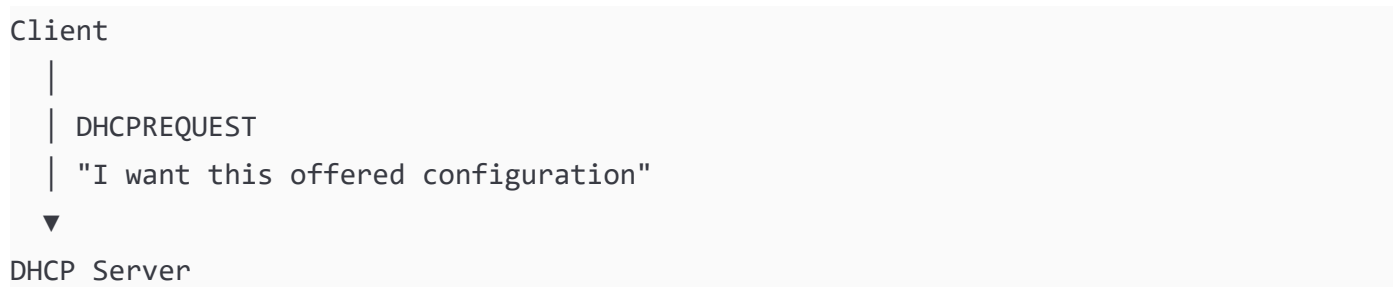
Therefore, the server can offer an available address from this pool.



The client may receive offers from multiple DHCP servers and selects an offer according to its selection process.

2.5 DHCPREQUEST — Request

After receiving an offer, the client sends a **DHCPREQUEST** message to formally request the offered configuration.



The request identifies the selected DHCP server and the configuration the client is requesting.

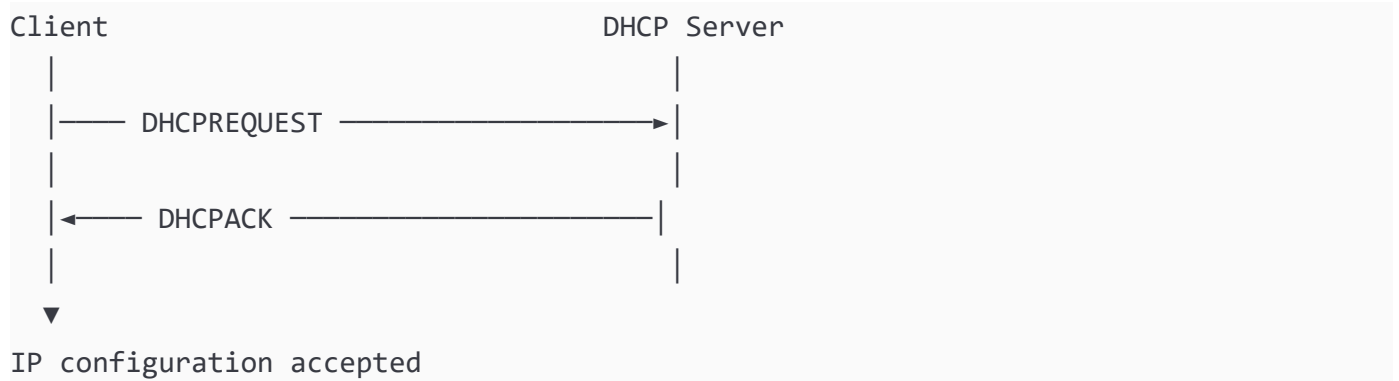
If multiple DHCP servers made offers, the request also indicates which offer was selected, allowing the other DHCP servers to withdraw their offers.

2.6 DHCPACK — Acknowledgment

The selected DHCP server receives the **DHCPREQUEST** and verifies the request.

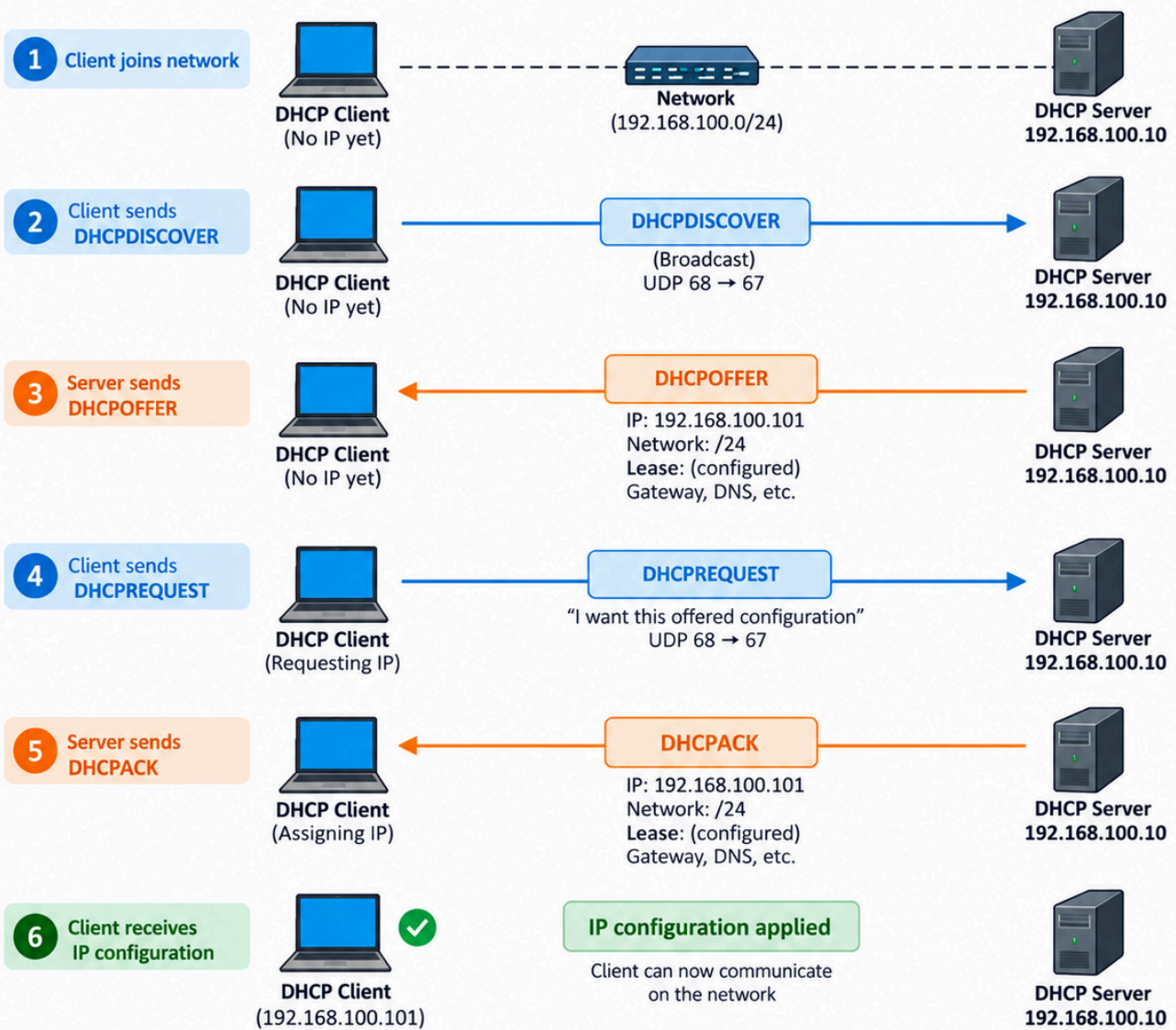
The server uses the transaction information, including the **XID (Transaction ID)**, to associate the request with the DHCP exchange.

If the request is valid, the server sends a **DHCPACK**.

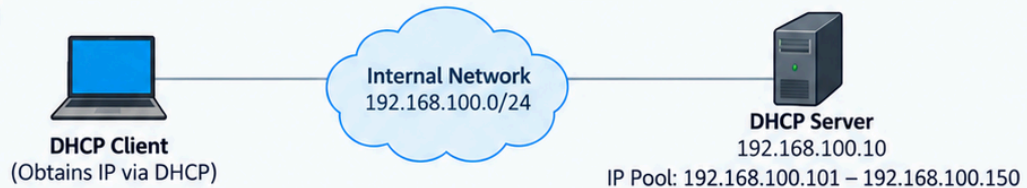


The DHCPACK confirms that the IP address and other network configuration settings have been assigned to the client.

2.7 Complete DHCP Flow



Lab Overview



This DORA process is the mechanism that will occur when our client is connected to the **Internal Network** and configured to obtain its IP address through DHCP.

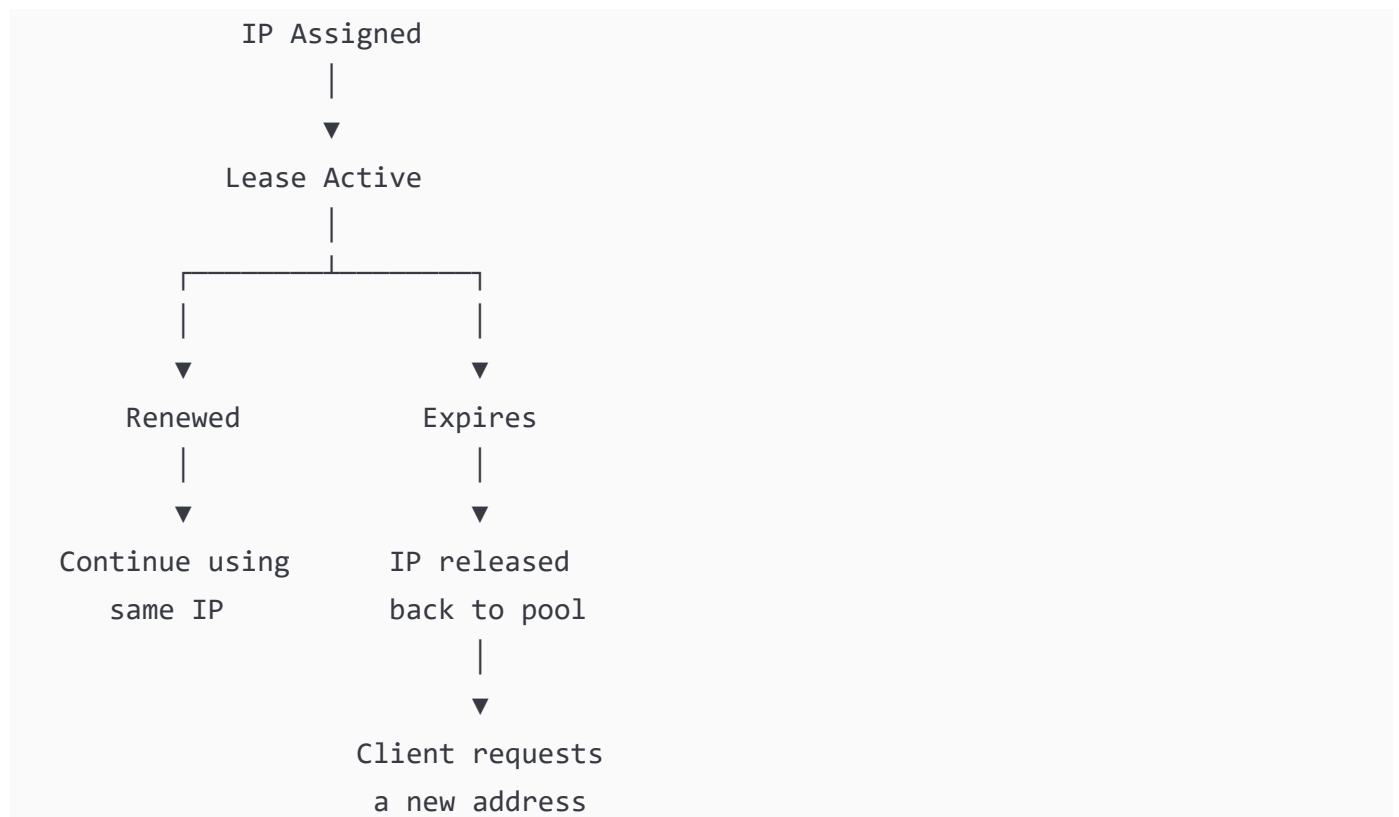
Section 3 — DHCP Lease Management

3.1 DHCP Lease

A DHCP server assigns an IP address to a client for a specific period called a **lease**.

The lease defines how long the client can use the assigned IP address.

Lease Lifecycle



Lease Management

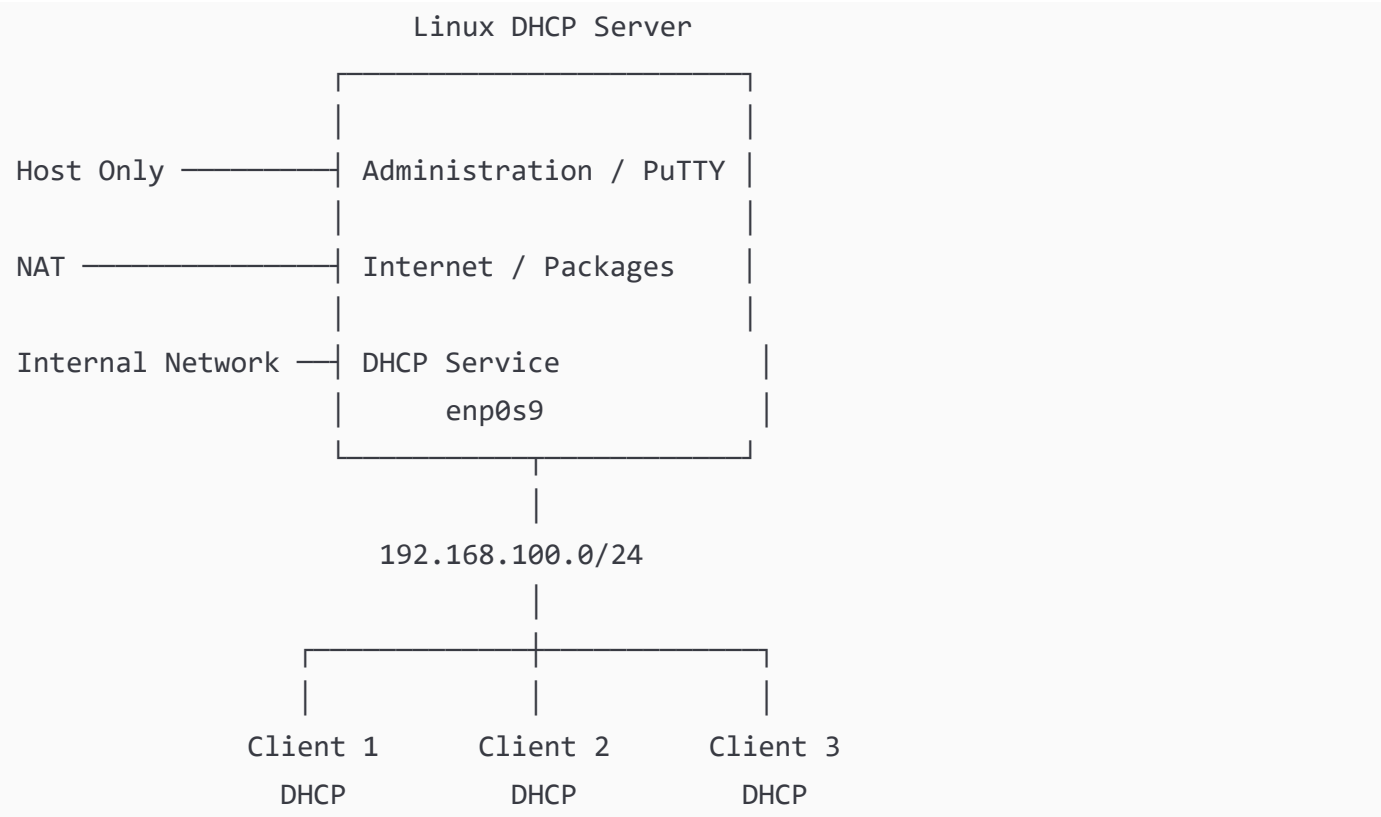
- **Lease Renewal:** As the lease approaches expiration, the client can request renewal. If approved, the client continues using the IP address.
- **Lease Expiration:** If the lease expires without renewal, the IP address is released and becomes available in the DHCP pool.
- **Client Release:** A client can voluntarily release its IP address when it no longer needs it.
- **DHCP Decline:** If the client detects that an offered IP address is already in use, it can send a **DHCPDECLINE** to the server.

This lease management process allows the DHCP server to track assigned addresses and return unused addresses to the available pool.

Section 4 — Lab Topology & Network Plan

4.1 Lab Network Topology

Our DHCP lab uses three network adapters on the Linux server:



Network Adapter	Purpose
Host Only	Server administration through PuTTY
NAT	Internet access and package installation
Internal Network	DHCP service and client communication

The DHCP service runs through the **Internal Network interface** `enp0s9`.

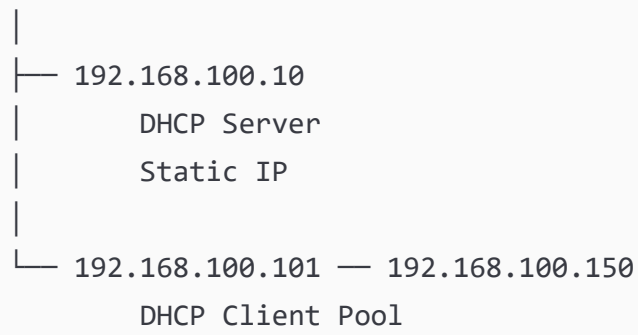
4.2 Network Plan

The DHCP server uses the following network:

Network:	192.168.100.0/24
Server IP:	192.168.100.10
DHCP Range:	192.168.100.101 – 192.168.100.150
Interface:	enp0s9

The server uses `192.168.100.10` as its **static IP**, while addresses from `192.168.100.101` to `192.168.100.150` are available for DHCP clients.

192.168.100.0/24



Section 5 — DHCP Server Installation & Configuration

5.1 Install DHCP Server

Update the package information and install the **ISC DHCP Server** package:

```
sudo apt update
sudo apt install isc-dhcp-server
```

The package used in this setup is:

```
isc-dhcp-server
```

5.2 Identify the Network Interface

Before configuring DHCP, identify the available network interfaces:

```
ip a
```

For our lab, the DHCP service runs on:

```
enp0s9
```

This is the interface connected to the **Internal Network**.

5.3 Configure the DHCP Interface

Edit the DHCP service's default configuration:

```
sudo nano /etc/default/isc-dhcp-server
```

Find:

```
INTERFACESv4=""
```

Set the DHCP interface:

```
INTERFACESv4="enp0s9"
```

This tells the DHCP server which network interface should handle DHCP requests.

5.4 Configure the DHCP Scope

Edit the main DHCP configuration file:

```
sudo nano /etc/dhcp/dhcpd.conf
```

Add the DHCP configuration:

```
authoritative;

subnet 192.168.100.0 netmask 255.255.255.0 {
    range 192.168.100.101 192.168.100.150;
}
```

Configuration

Directive	Purpose
<code>authoritative;</code>	Marks the DHCP server as authoritative for the network
<code>subnet</code>	Defines the network managed by DHCP
<code>range</code>	Defines the IP addresses available for clients

For this lab:

DHCP Server → 192.168.100.10

Client Pool → 192.168.100.101 – 192.168.100.150

The server's static address is kept outside the DHCP pool.

5.5 Validate the Configuration

Before starting the service, check the DHCP configuration for syntax errors:

```
sudo dhcpcd -t -cf /etc/dhcp/dhcpd.conf
```

If there are no configuration errors, proceed with the service.

5.6 Start the DHCP Service

Restart the DHCP service:

```
sudo systemctl restart isc-dhcp-server
```

Enable it to start automatically after boot:

```
sudo systemctl enable isc-dhcp-server
```

Section 6 — DHCP Service Verification & Status

After configuring the DHCP server, verify that the configuration is valid and that the `isc-dhcp-server` service is running correctly.

6.1 Check Configuration Syntax

Before restarting the service, test the DHCP configuration:

```
sudo dhcpcd -t -cf /etc/dhcp/dhcpd.conf
```

If there are no configuration errors, proceed with the service restart.

6.2 Check DHCP Service Status

Restart the DHCP service:

```
sudo systemctl restart isc-dhcp-server
```

Check its current status:

```
sudo systemctl status isc-dhcp-server
```

A successful setup should show the service as:

```
Active: active (running)
```

6.3 Verify DHCP Interface

Confirm that the DHCP server is listening on the intended interface:

```
ip addr show
```

For our lab, the DHCP service should use:

```
enp0s9
```

The interface is specified in:

```
/etc/default/isc-dhcp-server
```

with:

```
INTERFACESv4="enp0s9"
```

6.4 Check DHCP Listening Port

DHCP uses **UDP port 67** on the server. The listening socket can be checked with:

```
sudo ss -ltnp
```

Look for the DHCP server listening on UDP port `67`.

6.5 Check DHCP Logs

DHCP service messages can be checked from the system log:

```
sudo cat /var/log/syslog | grep dhcpcd
```

For live monitoring while testing:

```
sudo tail -f /var/log/syslog | grep dhcpcd
```

These logs help identify configuration, interface, or client-request problems.

Section 7 — DHCP Advanced Configuration

7.1 Lease & Server Settings

Additional global settings can be defined in `/etc/dhcp/dhcpd.conf`:

```
authoritative;

default-lease-time 86400;
max-lease-time 172800;

ddns-update-style none;
```

Setting	Purpose
<code>authoritative</code>	Identifies this DHCP server as authoritative for the network
<code>default-lease-time</code>	Default lease duration
<code>max-lease-time</code>	Maximum lease duration
<code>ddns-update-style none</code>	Disables automatic Dynamic DNS updates

7.2 Additional DHCP Options

DHCP can provide clients with additional network parameters such as:

```
option domain-name-servers <DNS-server>;
option domain-name "<domain-name>";
option routers <gateway>;
option broadcast-address 192.168.100.255;
```

These options provide **DNS, domain, gateway, and broadcast information** to clients.

7.3 DHCP Reservation

A specific client can receive a fixed IP based on its MAC address:

```
host client1 {
    hardware ethernet <MAC-address>;
    fixed-address <IP-address>;
}
```

This is useful when a particular client must consistently receive the same IP address.

Apply Changes

After modifying the configuration:

```
sudo dhcpd -t -cf /etc/dhcp/dhcpd.conf  
sudo systemctl restart isc-dhcp-server
```

Section 8 — DHCP Troubleshooting

Problem / Mistake	What to Check	Corrective Action
DHCP service fails to start	<code>dhcpd.conf</code> syntax and subnet configuration	Run <code>dhcpd -t</code> and fix reported errors
Client gets no IP	Client and server are on the same Internal Network	Check the VM network adapter settings
Wrong interface configured	<code>/etc/default/isc-dhcp-server</code>	Ensure <code>INTERFACESv4</code> points to the DHCP interface
Wrong subnet or IP range	<code>dhcpd.conf</code>	Ensure the subnet and <code>range</code> match the lab network
IP pool exhausted	<code>/var/lib/dhcp/dhcpd.leases</code>	Check existing leases and available addresses
DHCP service running but no response	UDP port <code>67</code> and DHCP logs	Check whether <code>dhcpd</code> is listening and review logs
Client receives unexpected configuration	DHCP options and multiple DHCP servers	Check for another DHCP server on the same network
Configuration changes have no effect	Service was not restarted	Validate the configuration and restart the DHCP service

Useful Commands

```
sudo dhcpd -t -cf /etc/dhcp/dhcpd.conf
sudo systemctl status isc-dhcp-server
sudo tail -f /var/log/syslog | grep dhcpd
sudo cat /var/lib/dhcp/dhcpd.leases
sudo ss -lunp
```

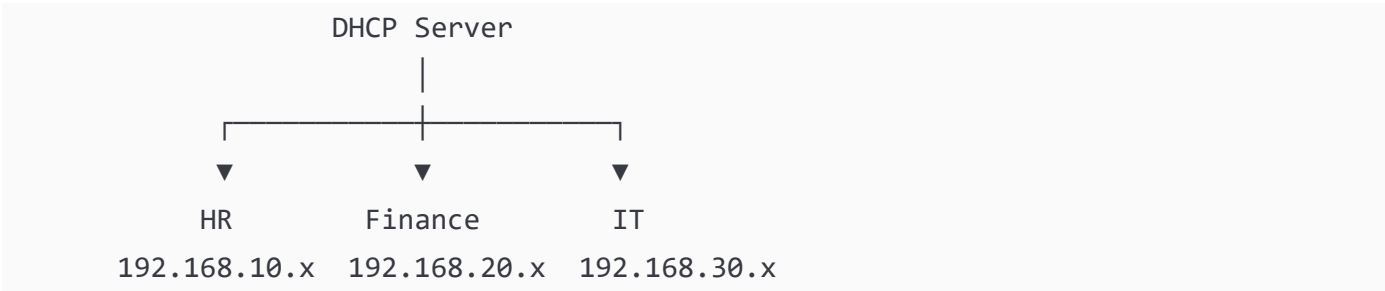
These checks help identify whether the problem is with the **network connection, interface, configuration, service, or DHCP lease**.

Section 9 — Real-World DHCP Scenario

9.1 Multi-Department Office

An organization has three departments, each using a separate subnet and DHCP pool.

Department	Network	DHCP Pool
HR	192.168.10.0/24	192.168.10.101-150
Finance	192.168.20.0/24	192.168.20.101-150
IT	192.168.30.0/24	192.168.30.101-150



A client receives an IP from the **pool belonging to its network**. Since each department has a different subnet, their DHCP pools do not overlap.

In an enterprise environment, these departments can be separated using **VLANs**, with a **DHCP relay** forwarding requests to a central DHCP server.

Section 10 — DHCP Viva Questions

Basic

1. **What is DHCP?**

Automatically provides IP addresses and network configuration to clients.

2. **What are DHCP's ports?**

UDP **67** — Server, UDP **68** — Client.

3. **What is DORA?**

Discover → Offer → Request → Acknowledgment.

4. **What is a DHCP lease?**

Temporary permission to use an assigned IP address.

5. **What is DHCP reservation?**

Assigning a fixed IP to a client based on its MAC address.

Linux

6. **Which DHCP package is used?**

```
isc-dhcp-server
```

7. **Main DHCP configuration file?**

```
/etc/dhcp/dhcpd.conf
```

8. **How do you test configuration?**

```
sudo dhcpd -t -cf /etc/dhcp/dhcpd.conf
```

9. **Where are DHCP leases stored?**

```
/var/lib/dhcp/dhcpd.leases
```

10. **How do you check the DHCP service?**

```
sudo systemctl status isc-dhcp-server
```

Troubleshooting

11. **Client gets no IP — what do you check?**

Network connection, DHCP interface, subnet/range, service, and logs.

12. **Why might a client get an unexpected IP?**

Incorrect configuration or another DHCP server on the network.

13. **What happens when the DHCP pool is exhausted?**

The server cannot assign an available address to a new client.